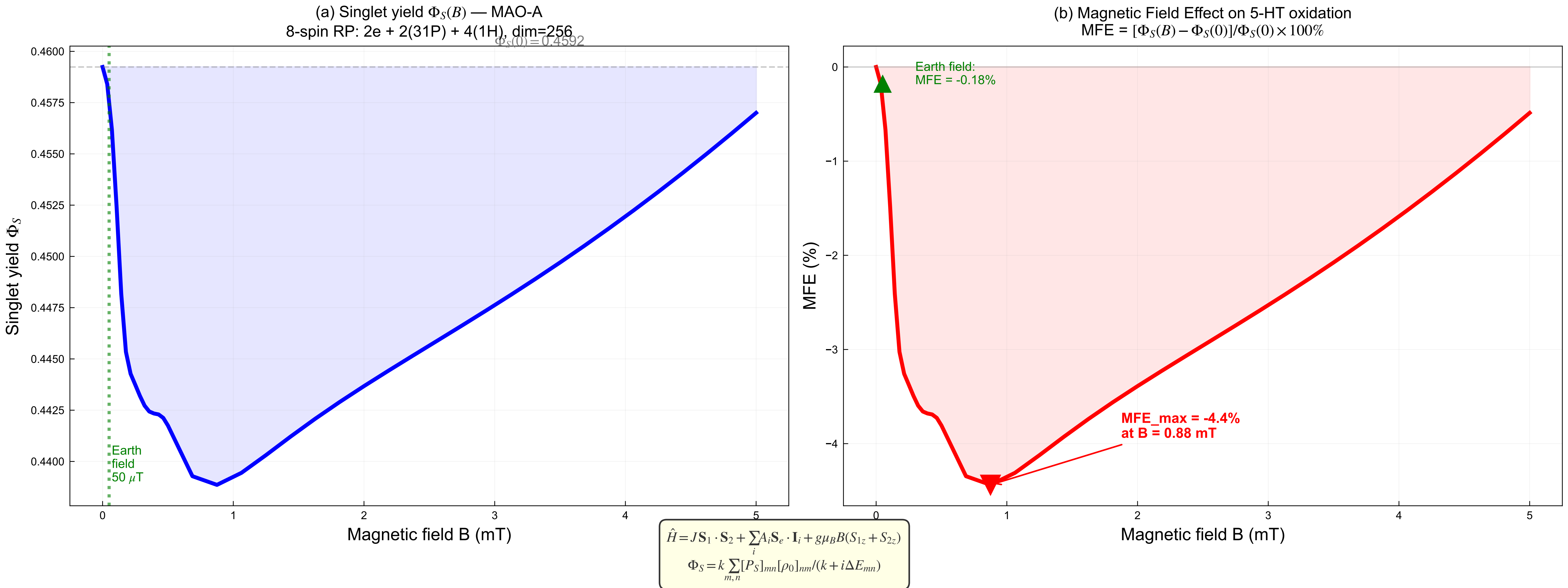

(a) Singlet yield $\Phi_S(B)$ — MAO-A
8-spin RP: 2e + 2(31P) + 4(1H), dim=256
$\Phi_S(0) = 0.4592$
Singlet yield Φ_S
Magnetic field B (mT)
Earth field 50 μT
0.4600
0.4575
0.4550
0.4525
0.4500
0.4475
0.4450
0.4425
0.4400
0
1
2
3
4
5
(b) Magnetic Field Effect on 5-HT oxidation
MFE = $[\Phi_S(B) - \Phi_S(0)]/\Phi_S(0) \times 100\%$
MFE (%)
Magnetic field B (mT)
Earth field: MFE = -0.18%
MFE_max = -4.4% at B = 0.88 mT
0
−1
−2
−3
−4
0
1
2
3
4
5
$\hat{H} = J\mathbf{S}_1 \cdot \mathbf{S}_2 + \sum_i A_i \mathbf{S}_e \cdot \mathbf{I}_i + g\mu_B B(S_{1z} + S_{2z})$
$\Phi_S = k \sum_{m,n} [P_S]_{mn}[\rho_0]_{nm}/(k + i\Delta E_{mn})$